4G Beidou intelligent safety helmet
TK-SURY01
Product white paper

1. product description

The TK-SURY01 is a smart safety helmet independently developed by Cloudwinner Network, featuring 4G HD video transmission, monitoring alerts, team intercoms, and BeiDou positioning. Utilizing mobile internet, IoT, sensor technology, high-definition wide-angle cameras, and AI, it offers voice control, real-time video streaming, photo/video recording, audio capture, instant communication, GPS tracking, electronic fencing, safety alerts, facial recognition, and video behavior analysis. Optional features include proximity detection, gas monitoring, and vital sign detection. Data supports both real-time uploads and offline storage. Through integration with converged communication platforms, it enables remote operations, expert guidance, and AI-powered on-site video analysis for environmental monitoring and defect detection. This system delivers comprehensive solutions for field operations, safety management, defect identification, and risk alerts across various industrial scenarios.

2. Product diagram

TK-SURY01 appearance diagram

3. product function

1、Network communication: support 4G, WiFi5, Bluetooth 4.2 communication;

2、Video function: support high-definition photography, 1080P local recording, 1080P HD video back transmission, video call and other functions;

3、Voice function: support single voice intercom, communication group intercom, PTT intercom, voice broadcast and other functions; support local key voice hangup;

4、Voice broadcast: supports voice broadcast functions such as platform message delivery, group message delivery and broadcast of security matters;

5、Outdoor positioning: It supports Beidou dual-frequency positioning, and combines multiple fusion positioning methods such as 4G base station and wireless network to achieve fast and accurate positioning. The positioning accuracy is within 3 meters in open outdoor areas and within 10 meters in urban high-rise environment;

6、Indoor positioning: embedded Bluetooth positioning program, support for Bluetooth beacon positioning, bluetooth beacon positioning accuracy 3-6 meters;

7、Emergency Alert: The smart safety helmet features a prominent SOS button on the exterior. Even when operators wear gloves or operate in dark, obstructed environments, the device can quickly detect triggers through blind touch. The button includes anti-misoperation protection, requiring a 3-second long press to activate, ensuring 100% alarm success rate. Management Platform Alert: Upon receiving an SOS signal, the platform automatically displays an alert window with audio notification, showing the rescuer's identity, location coordinates, and trigger time. The system also marks the exact location on an electronic map, enabling management to swiftly formulate rescue plans.

8、Alarm function: support off hat alarm, fall alarm, collision alarm, silent alarm, high altitude alarm, near electricity alarm (optional), toxic gas monitoring and early warning (optional), vital sign monitoring and early warning (optional);

9、Speech recognition: Support voice command set and natural language interaction to control the hat for corresponding operations, such as equipment control,

information query, work coordination, etc., so as to free hands and ensure efficient operation;

10、Electronic Fence: The platform enables the establishment of electronic fence zones. When personnel wearing smart safety helmets cross the boundary, the helmet triggers an alarm through high-pitched voice announcements locally while simultaneously sending real-time intrusion data to the control platform, including the wearer's identity, time of crossing, and precise coordinates. The system supports various alarm types such as entry prohibition, exit prohibition, unauthorized departure, overcapacity alerts, and lingering status notifications.

11、Personnel attendance management: The platform sets up an electronic fence area. When the personnel wearing safety helmet enters or leaves the specified area, the attendance is automatically recorded to achieve refined attendance management;

12、Battery life: The detachable battery is standard with a large 5000mAh detachable battery, and the continuous video recording life can reach more than 8 hours;

13、AI recognition function: support speech recognition/voice command control, natural voice interaction (optional), face recognition (optional), behavior recognition (optional);

14、Harmful gas monitoring (optional): with a gas detector, it provides abnormal alarm for the concentration of oxygen, carbon monoxide, methane, hydrogen sulfide, ammonia, carbon dioxide and other gases;

15、Life sign monitoring (optional): equipped with a Bluetooth wristband, providing abnormal alarm of personnel's heart rate, blood oxygen, blood pressure, etc.;

16、Remote configuration: support remote configuration and management through the APP tool, remote function configuration, remote restart, WiFi configuration, software upgrade, etc.;

17、Protection level: IP66, with waterproof, dustproof, shock resistance and other characteristics, adapt to a variety of harsh working environment;

18、Product certification: passed GB2811-2019 "Head Protection Safety helmet" inspection and testing, Beidou independent positioning test certification, explosion-proof certification.

4. Product Parameter

type	project	description
Surface	Pigment	White, red, blue, yellow
	Characteristic	● Mark the printing area in front ● Reserve a printing area for side identification ● The rear reserved marking printing area
Size and weight	Material quality	ABS material quality
	Size	287mm*217mm*156mm (length * width * height) "Meet the design of GB/T2811-2019 standard"
	Weight	680g
Material and protective performance	Shock absorption	Designed according to GB/T2811-2019 standard
	Puncture resistant	Designed according to GB/T2811-2019 standard
	IP grade	IP66
	Falling prevention	If you fall from 2 meters, the function will fail
Computing and storage	CPU	Arm Cortex A53
	RAM	2GB
	Flash	16GB
	System	Android 10
Communication and positioning	Communication frequency band	TDD-LTE: B34/38/39/40/41 FDD-LTE: B1/3/5/7/8/20 WCDMA: B1/B5/B8 GSM: B2/3/5/8
	Fixed position	Beidou dual-frequency positioning (optional RTK)
	WIFI	Support dual frequency (IEEE802.11a/b/g/n/ac)
	PDA	Supports Bluetooth 4.2 1. The safety helmet can be connected to the Bluetooth headset, and the headset can broadcast voice content and receive voice calls 2. The Bluetooth connection of the safety helmet can scan the Bluetooth beacon for positioning 3. Vital sign monitoring: Connect to the Bluetooth bracelet 4. Toxic gas monitoring: Connect the gas detector
Peripheral interface	Type-C joggle	Supports charging and data transmission
	SIM card slot	NANO SIM card (dual SIM)
	TF Card slot	Standard 64G, up to 256G support
Cell	Battery specifications	• capacity 5000mAh • Removable, battery cell rechargeable
Peripheral	Loudspeaker	8 Euro 1 watt single speaker
	Microphone	Configuration: main MIC, sub MIC (noise

		reduction), voice MIC
	LED front light	Support, 1W
	Laser light assists in pointing	support
	Camera	standard configuration • Pixel: 8 megapixels, 1080P • Horizontal Angle: 120 degrees
Sensor	Close range sensors acceleration transducer gyroscope barometer Proximity sensor	• Take off your hat and call the police • Fall alarm • Call for a collision • Silent alarm • Call 911 • Near field alarm (optional)
AI discriminate		• Voice recognition, voice control • Face recognition (optional) • Behavior recognition (optional)
Key	Key	Switch button: press and hold for 2 seconds to turn on or off Photo/video button: 1. Short press to take a photo in preview mode; 2. Long press for 2s to enable or disable video recording Light button: 1. Short press to turn on or off the LED white light in preview mode; 2. Long press for 2s to turn on or off the laser auxiliary lamp SOS button: press for 3 seconds to send an alarm signal to the platform PTT key: 1. Long press to carry out PTT intercom; 2. Short press to open speech recognition
Status indicator	Pilot lamp	The red light stays on during charging The green light is always on
Fittings of a machine	Source	Adapter 1, power 5V/2A
	Data wire	Type C, line length 1 meter
	Cell	5000mAh
	Lining	1 copy
Work environment	Working temperature	-10℃ ~ 60℃
	Working humidity	5% ~ 95%

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